

Tournament Selection

Choose the best from randomly selected K solutions.
(K : Tournament Size)

Iterate the following steps:

Step 1: Randomly select K solutions from the current population
(with duplication or without duplication).

Step 2: Select the best solution among the K solutions.

==> Selection probability of each solution depends on its rank (and the value of K).

Lab Session Task 1:

1. Calculate the selection probability of the best solution for the case of $N = 10$ (population size) and $K = 2$ (tournament size) “with duplication”.
2. Calculate the selection probability of the i -th worst solution for the case of $N = 10$ (population size) and $K = 2$ (tournament size) “with duplication”.
3. Calculate the selection probability of the i -th worst solution for the case of $N = 10$ (population size) and $K = 3$ (tournament size) “with duplication”.

Lab Session Task 2:

(1) In Simulated Annealing, clearly explain the advantages and the disadvantages of each of the following two strategies: (i) the temperature is gradually decreased from a high temperature to a low temperature, and (ii) the temperature is fixed at a pre-specified temperature (i.e., the initial temperature and the final temperature are the same).

(2) In Simulated Annealing, which do you think is easier between the following two tasks: (i) to appropriately specify the initial temperature, the final temperature, and the cooling schedule between the initial and final temperatures, and (ii) to appropriately specify the fixed temperature. Please clearly explain the reason for your answer.

Lab Session Task 3:

In evolutionary computation, which do you think the best generation update mechanism among the following mechanisms? Please clearly explain the reason for your answer.

(1 , 1)ES: Random search

(1 + 1)ES: Fast move local search

(1 + λ)ES: Local search (e.g., $\lambda = 100$)

(μ + 1)ES: Steady state algorithm (e.g., $\mu = 100$)

(μ , μ)ES: The simplest model of generation update

(μ + μ)ES: The standard plus strategy

(μ , λ)ES: The standard comma strategy (e.g., $\mu = 100$, $\lambda = 500$)